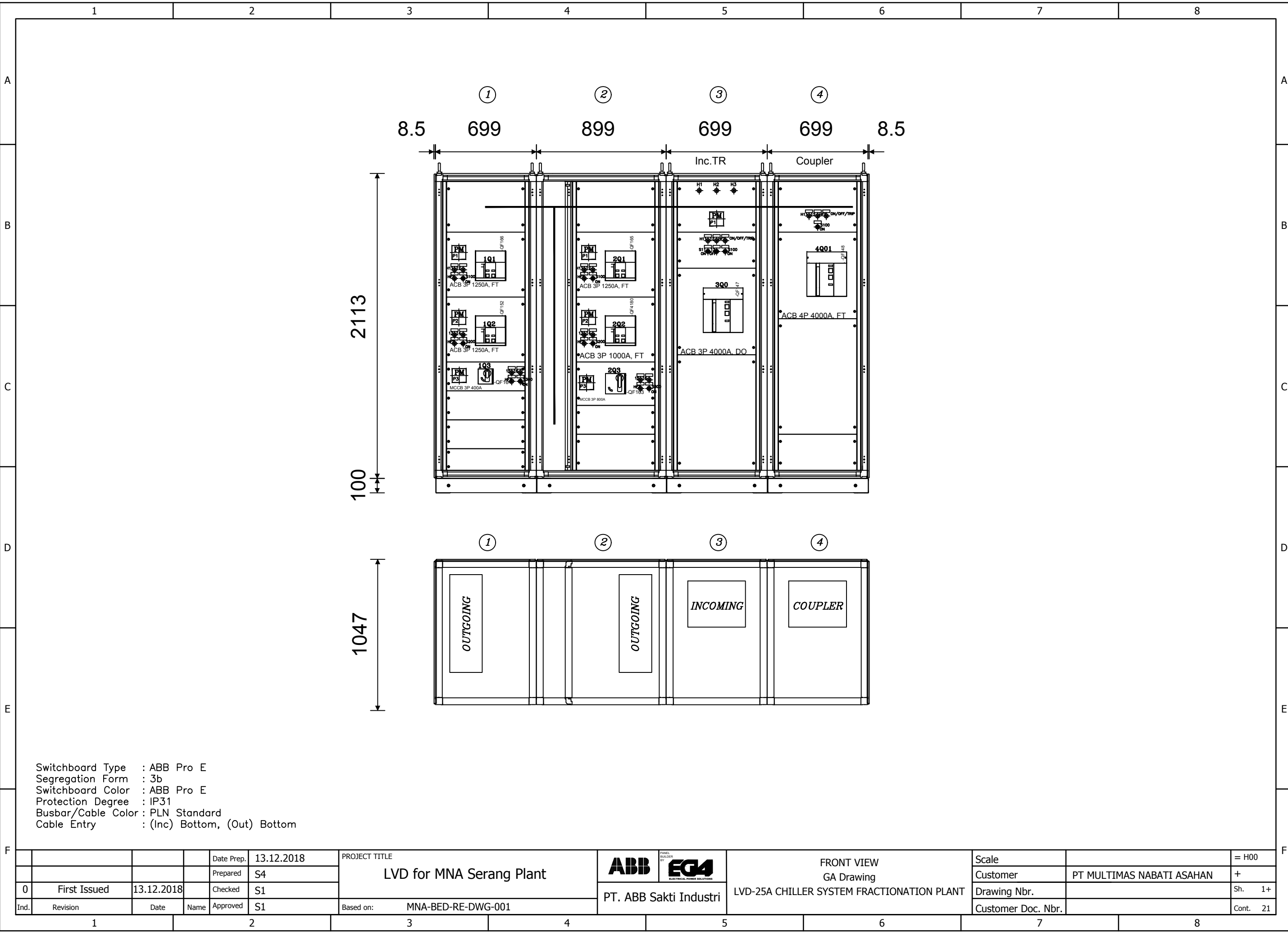
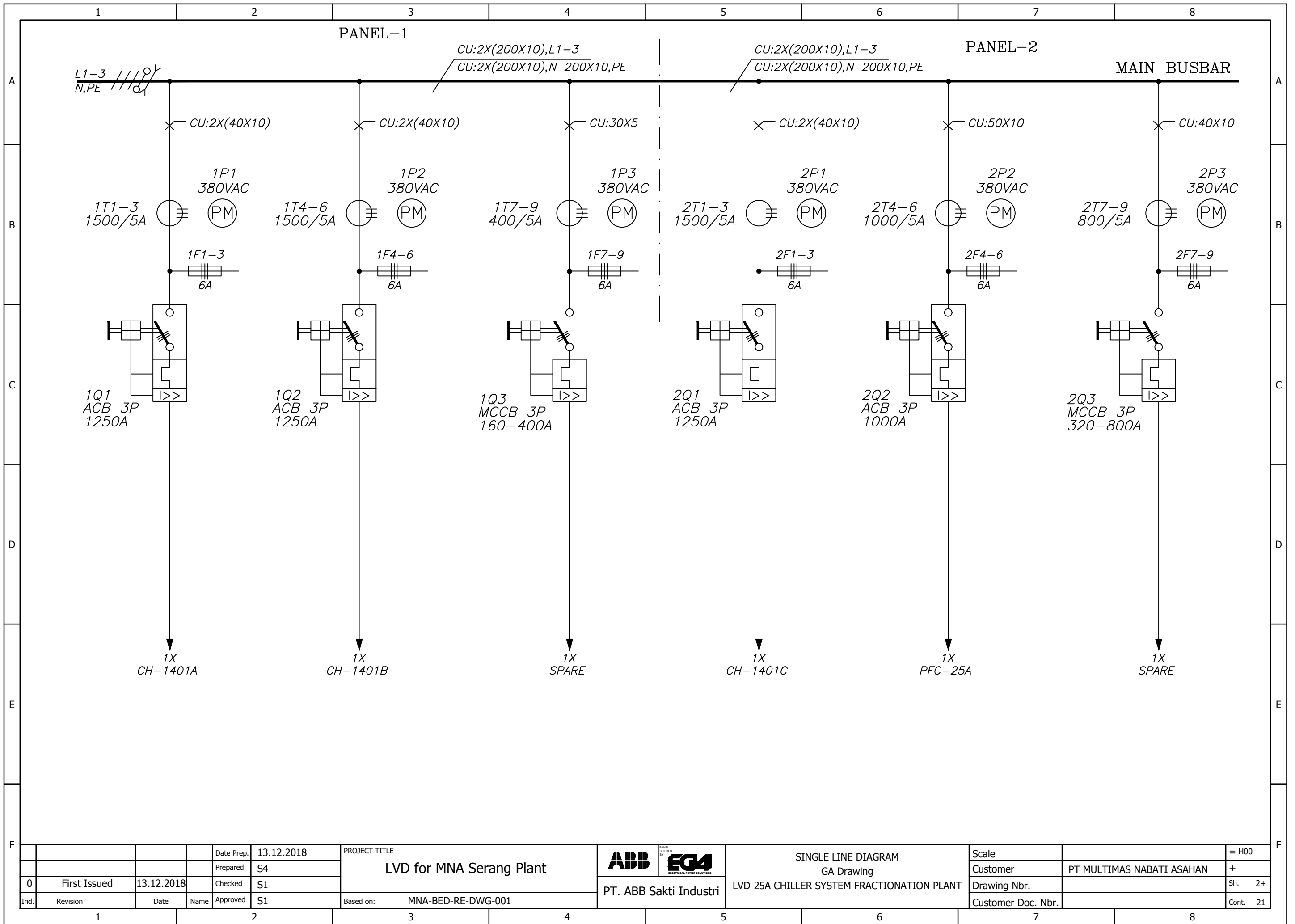


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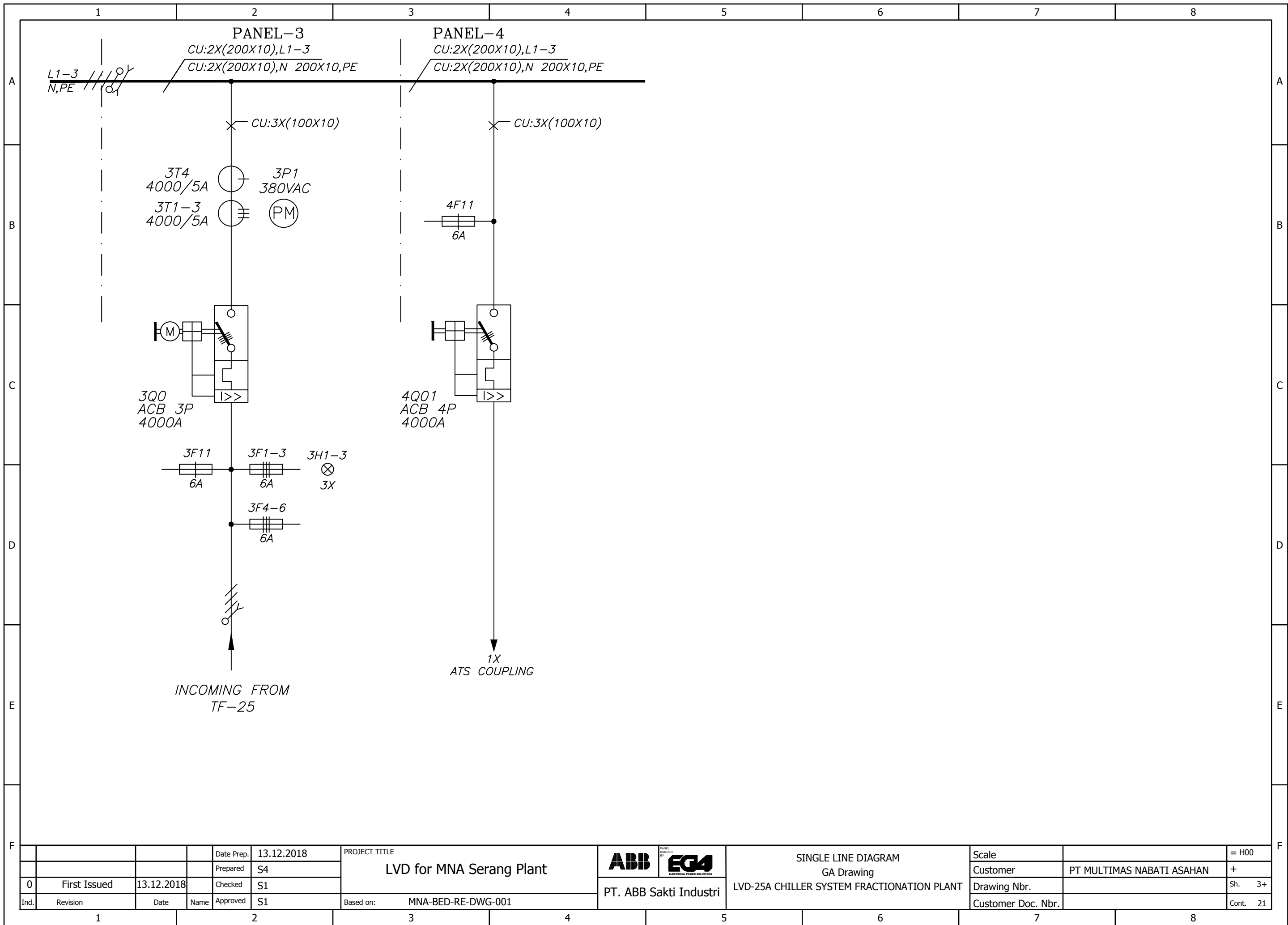


				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDERS BY EG4 ELECTRICAL POWER SOLUTIONS	FRONT VIEW GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT	Scale		= H00
			Prepared	S4	Customer				PT MULTIMAS NABATI ASAHAN	+	
0	First Issued	13.12.2018	Checked	S1	Drawing Nbr.					Sh. 1+	
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 21

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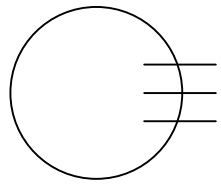


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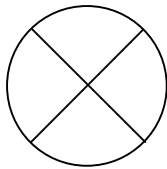


				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDER BY EG4 ELECTRICAL POWER SOLUTIONS PT. ABB Sakti Industri	SINGLE LINE DIAGRAM GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 3+
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

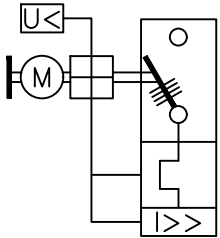
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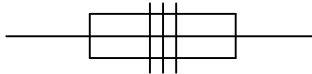
CT



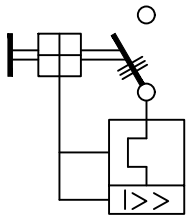
PILOT LAMP



ACB



FUSE



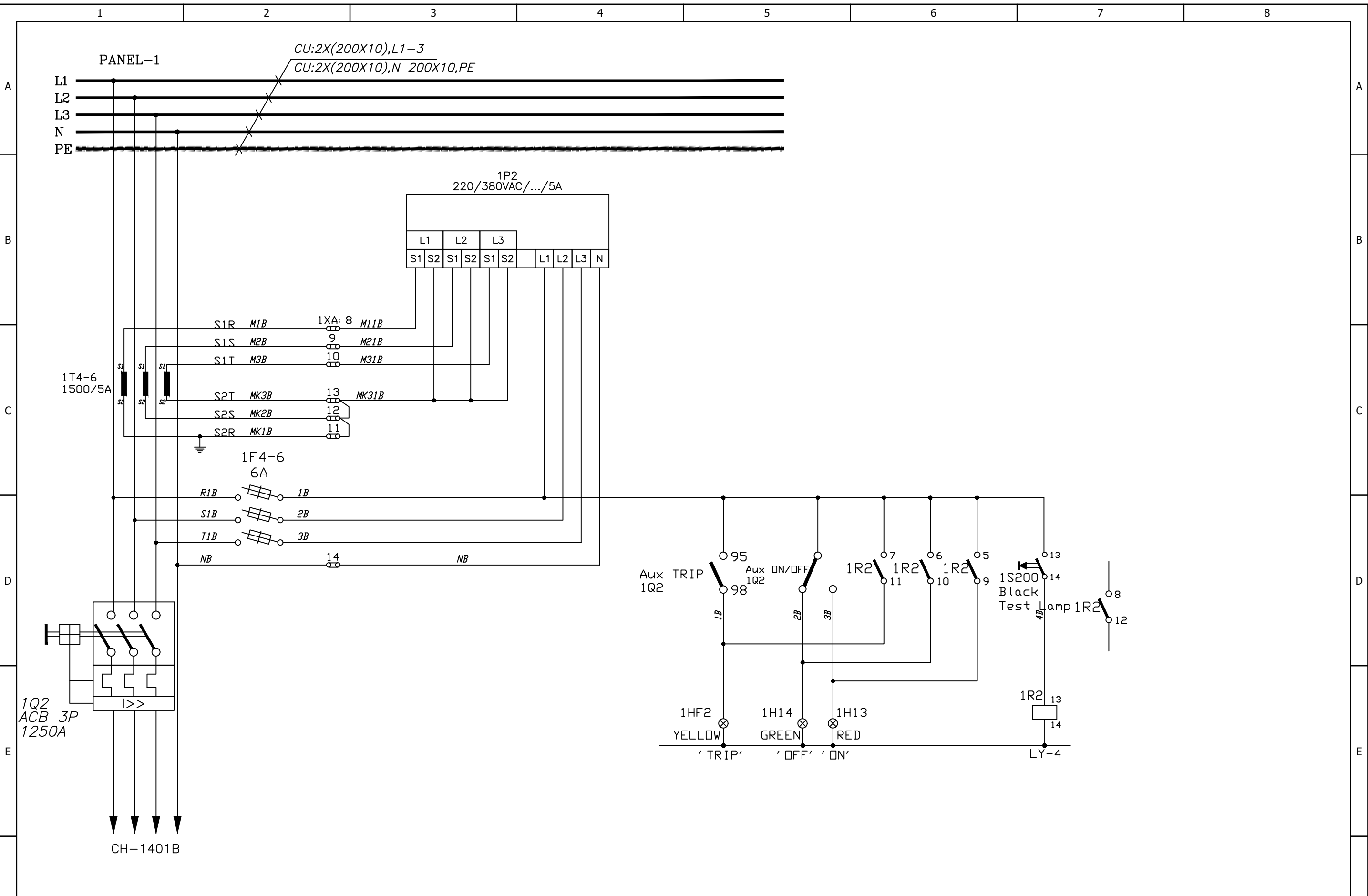
MCCB



DIGITAL POWER METER

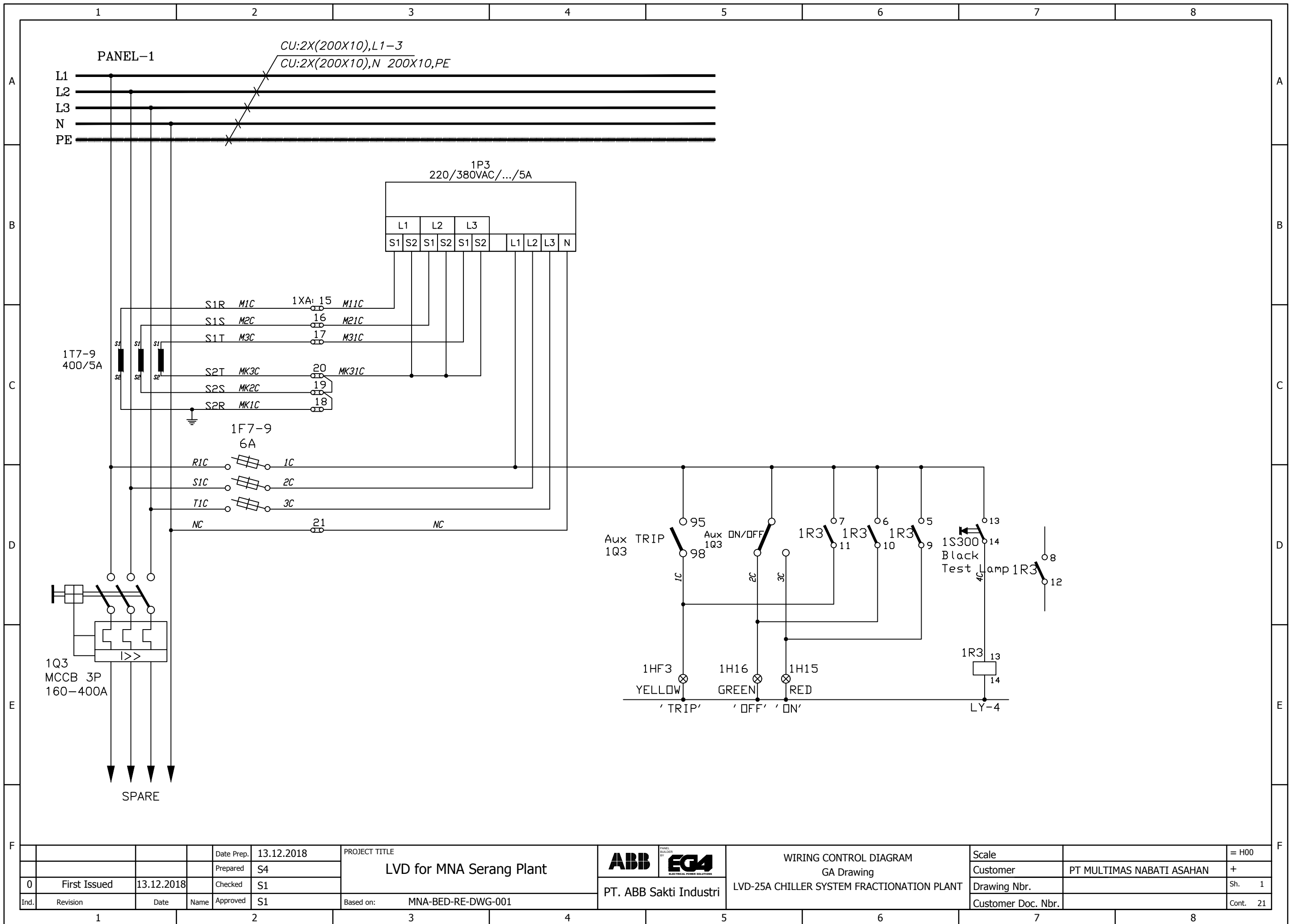
				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDER BY EG4 ELECTRICAL POWER SOLUTIONS PT. ABB Sakti Industri	LEGEND GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 4+
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

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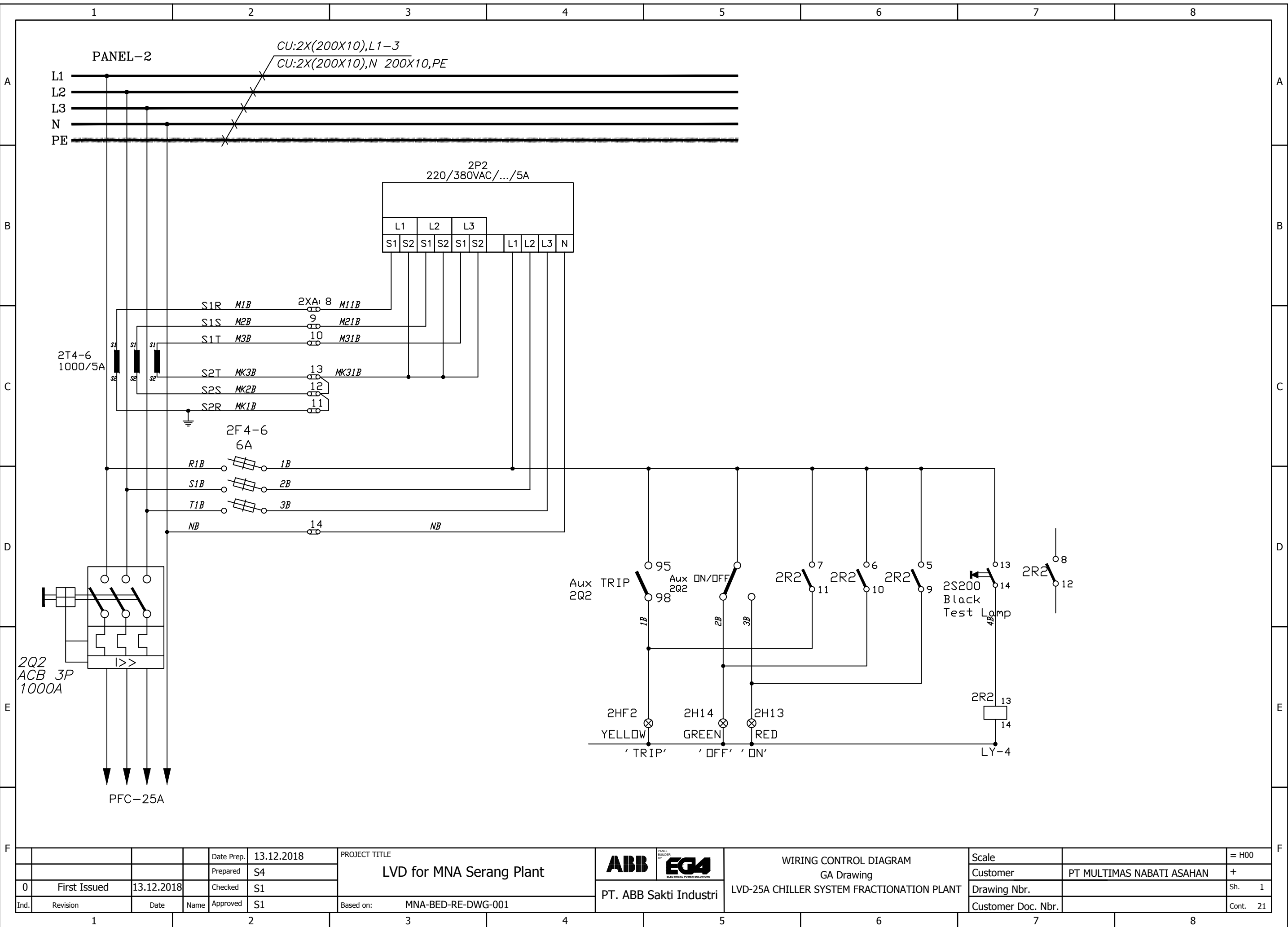


Ind.		Revision	Date	Name	Approved	Ind.	PROJECT TITLE		WIRING CONTROL DIAGRAM		Scale	= H00	
0		First Issued	13.12.2018		Checked	S1	LVD for MNA Serang Plant		GA Drawing		Customer	PT MULTIMAS NABATI ASAHAN	+
							Based on: MNA-BED-RE-DWG-001		LVD-25A CHILLER SYSTEM FRACTIONATION PLANT		Drawing Nbr.		Sh. 1
											Customer Doc. Nbr.		Cont. 21

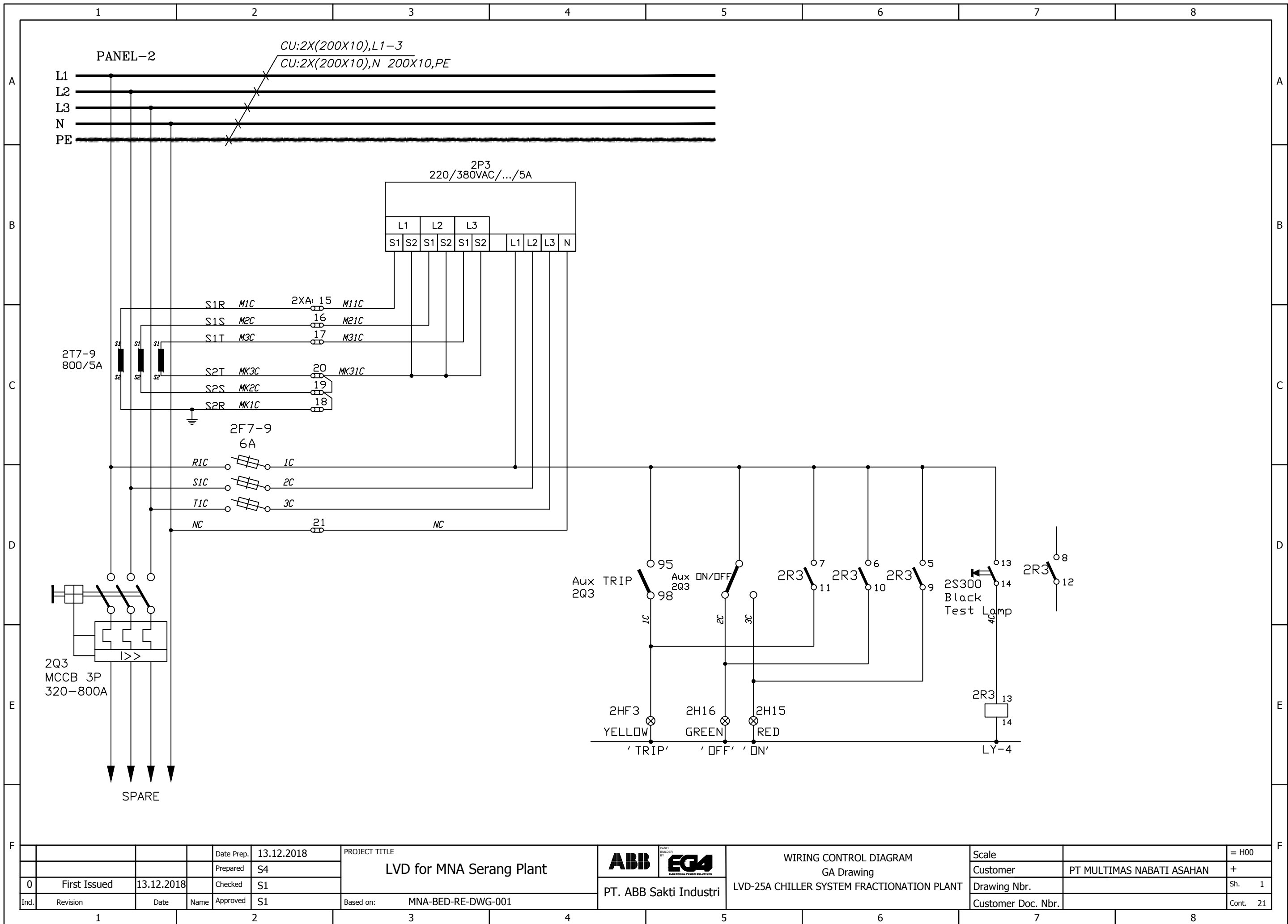
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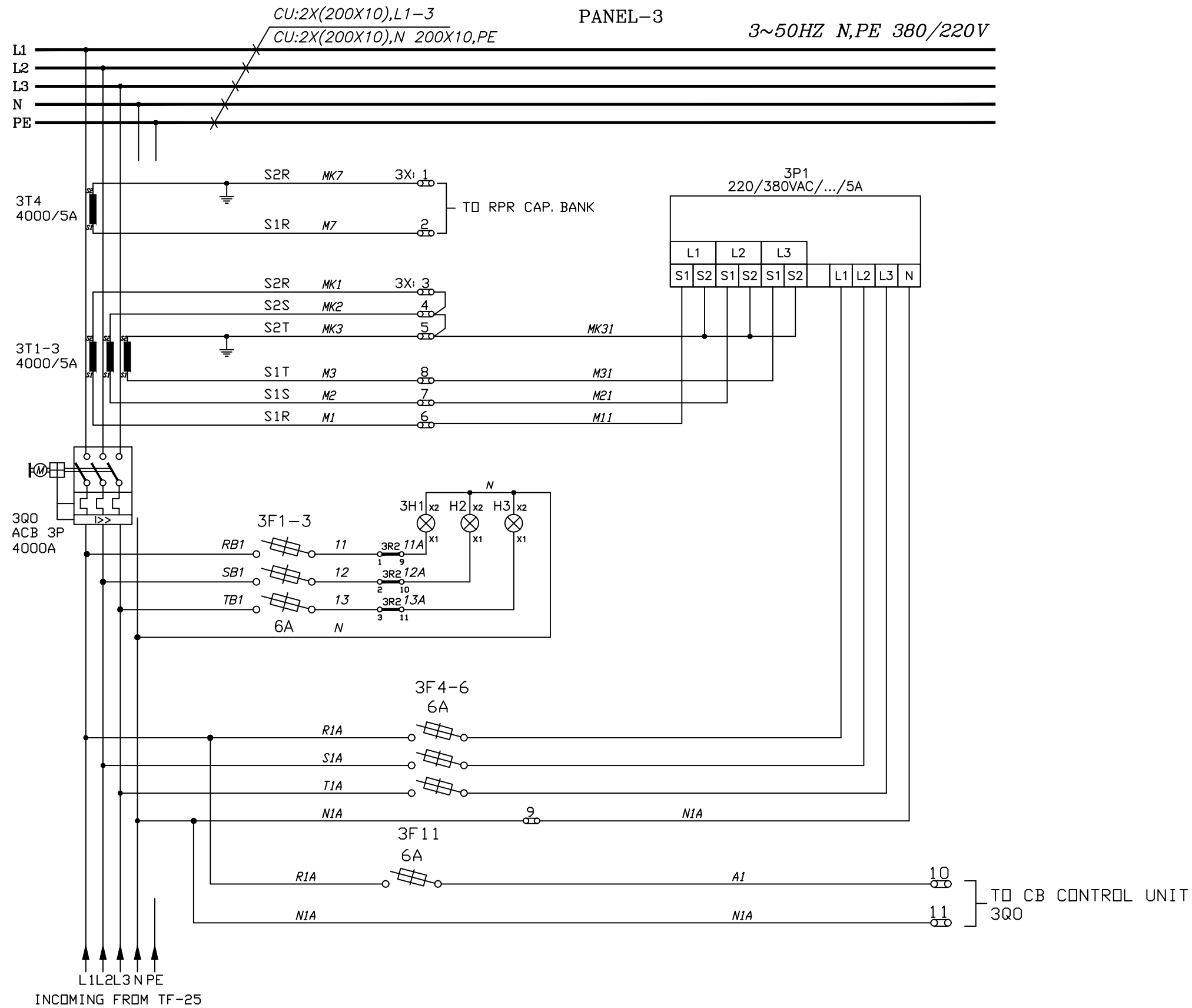
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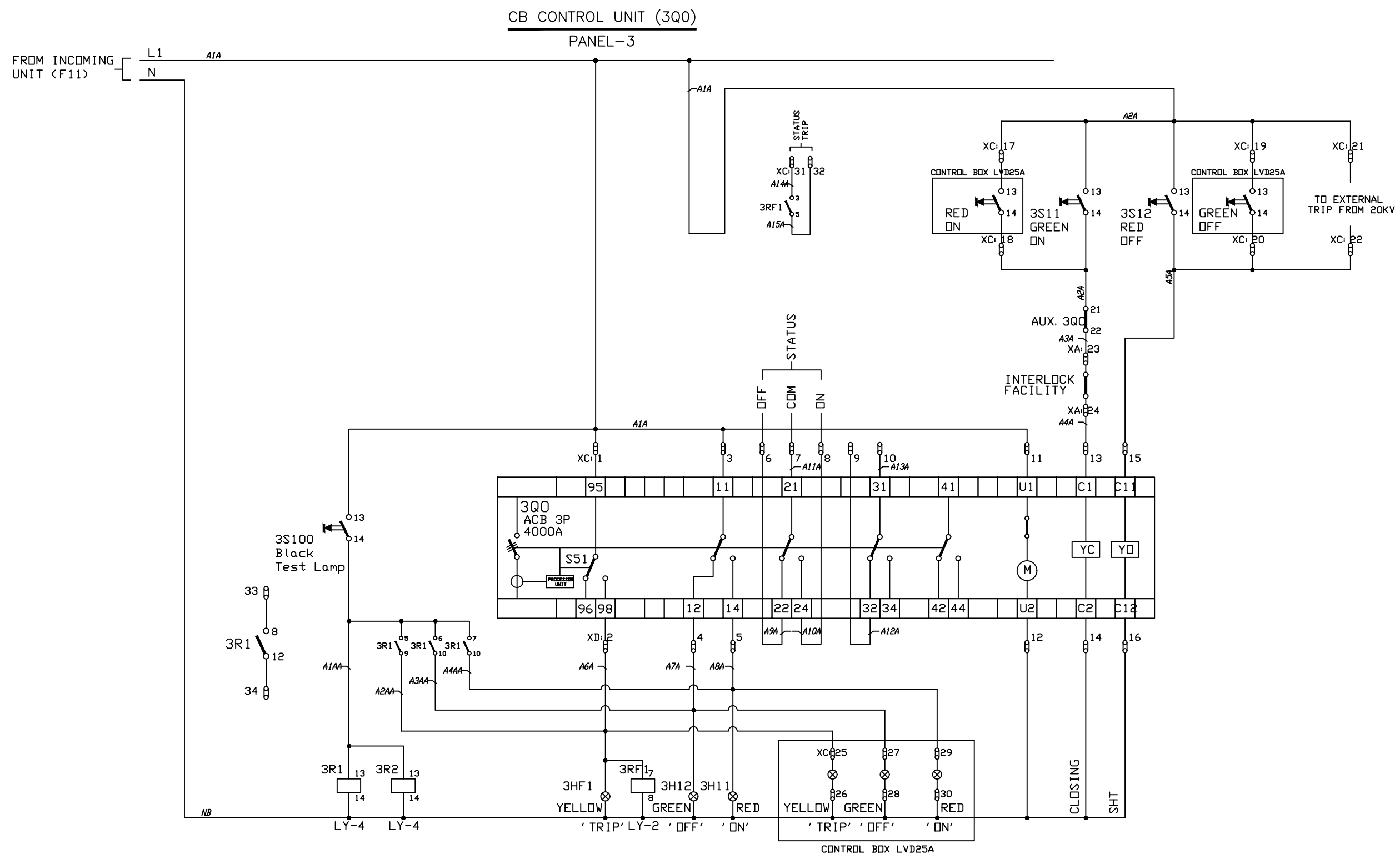


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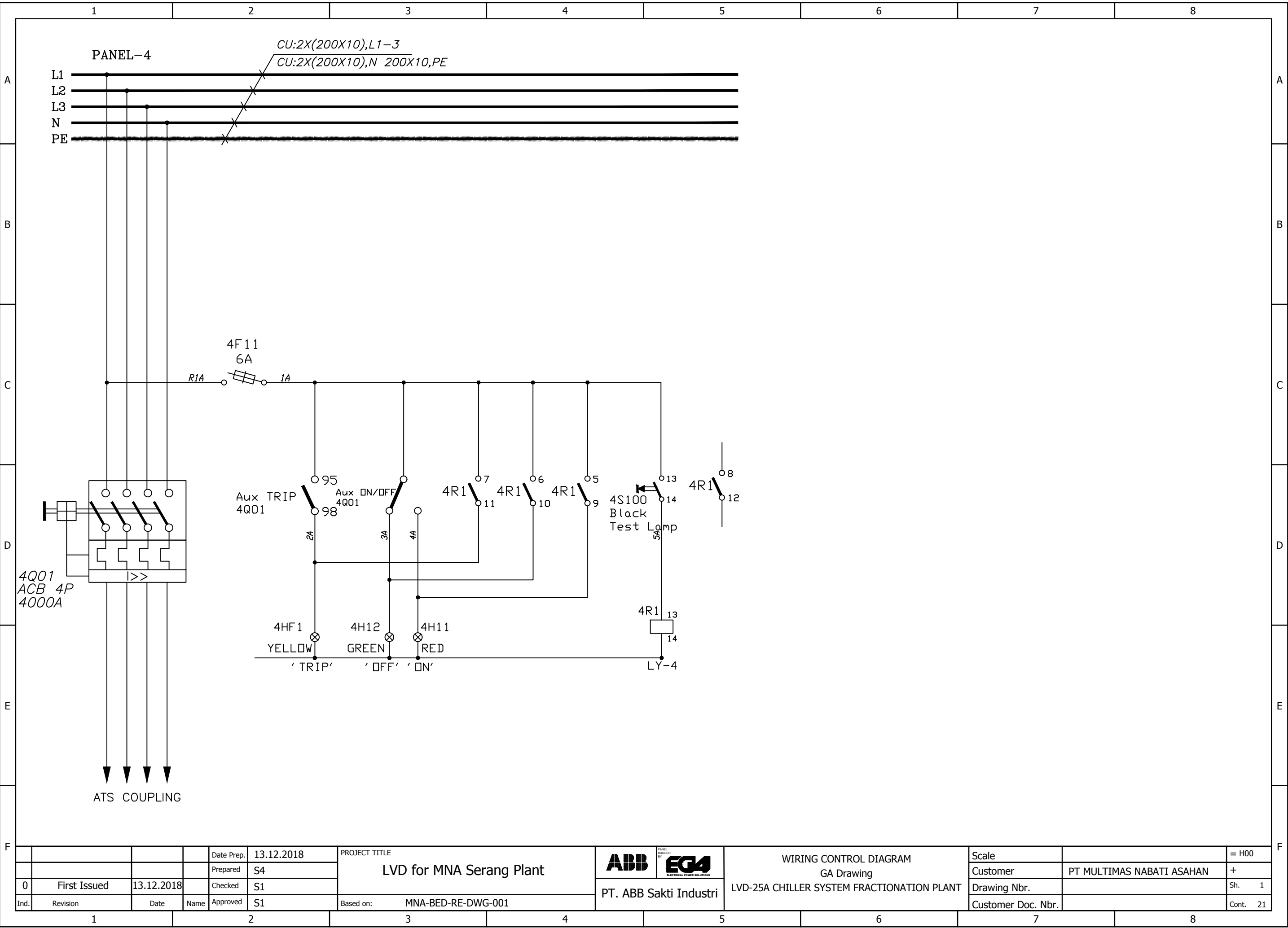
				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	  PT. ABB Sakti Industri	WIRING CONTROL DIAGRAM GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001		Customer Doc. Nbr.		Cont. 21

1 2 3 4 5 6 7 8



				Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant	ABB PANEL BUILDER BY EG4 ELECTRICAL POWER SOLUTIONS	WIRING CONTROL DIAGRAM GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT	Scale		= H00
				Prepared	S4				Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1				Drawing Nbr.		Sh. 1
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001	PT. ABB Sakti Industri	Customer Doc. Nbr.		Cont. 21	

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1		2		3		4		5		6		7		8	
A	N O	DESCRIPTION			TECHNICAL DATA			QTY	DESIGNATION						A
		INCOMING													
B	1	Fuse Carrier 6A			AXLE/Setara			7	3F11,3F1-6						B
	2	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			2	3H1,11						
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			2	3H2,HF1						
	4	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			2	3H3,12						
	5	DIGITAL POWER METER (*)			ABB, M2M ETHERNET			1	3P1						
	6	CT 4000/5A CL.0.5 40VA			Howig, HG 120-4000			4	3T1-4						
C	7	ACB 3P 4000A, 66KA, DO, Ekip Touch LSIG (*)			ABB, E4.2N40 D/O			1	3Q0						C
	8	+ Shunt Trip Release (Y0) (*)			ABB, Y0			1	-						
	9	+ Motor (M) + Closing Release (YC) For E2.2-E6.2 (*)			ABB, M+YC			1	-						
	10	+ Padlocks For E2.2-E6.2 (*)			ABB,			1	-						
	11	+ Ekip Com Protocol IEC 61850 (*)			ABB,			1	-						
	12	Push Button On, 1NO, Green			Ega			1	3S11						
	13	Push Button Off, 1NO, Red			Ega			1	3S12						
	14	Control Relay 220Vac			Omron, LY-2			1	3RF1						
	15	Control Relay 220Vac			Omron, LY-4			2	3R1-2						
	16	Push Button Test Lamp , 1NO, Black			Ega			1	3S100						
D	17														D
	18														
	19	COUPLER													
	20	Fuse Carrier 6A			AXLE/Setara			1	4F11						
E	21	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE/Setara			1	4H11						E
	22	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE/Setara			1	4HF1						
	23	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE/Setara			1	4H12						
	24	ACB 4P 4000A, 66KA, FT, Ekip Dip LI (*)			ABB, E4.2N40 F/T			1	4Q01						
	25	Push Button Test Lamp , 1NO, Black			Ega			1	4S100						
	26	Control Relay 220Vac			Omron, LY-4			1	4R1						
F					Date Prep.	13.12.2018	PROJECT TITLE LVD for MNA Serang Plant		  PT. ABB Sakti Industri	COMPONENT LIST GA Drawing LVD-25A CHILLER SYSTEM FRACTIONATION PLANT		Scale		= H00	F
					Prepared	S4						Customer	PT MULTIMAS NABATI ASAHAN	+	
	0	First Issued	13.12.2018		Checked	S1						Drawing Nbr.		Sh. A+	
	Ind.	Revision	Date	Name	Approved	S1	Based on:	MNA-BED-RE-DWG-001	Customer Doc. Nbr.		Cont. 21				
	1		2		3		4		5		6		7		

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1		2		3		4		5		6		7		8		
A	N O	DESCRIPTION			TECHNICAL DATA			QTY	DESIGNATION						A	
		OUTGOING														
B	1	Fuse Carrier 6A			AXLE			18	1,2F1-9						B	
	2	Pilot Lamp (LED) 220Vac/50Hz, Red			AXLE			6	1,2H11,13,15							
	3	Pilot Lamp (LED) 220Vac/50Hz, Yellow			AXLE			6	1,2HF1-3							
	4	Pilot Lamp (LED) 220Vac/50Hz, Green			AXLE			6	1,2H12,14,16							
	5	DIGITAL POWER METER (*)			ABB, M2M ETHERNET			6	1,2P1-3							
	6	CT 1500/5A CL.0.5 20VA			Howig, HG 82-1500			9	1T1-6,2T1-3							
	7	CT 1000/5A CL.0.5 10VA			Howig, HG 82-1000			3	2T4-6							
	8	CT 800/5A CL.1 10VA			Howig, HG 50I-800			3	2T7-9							
	9	CT 400/5A CL.1 5VA			Howig, HG 40I-400			3	1T7-9							
C	10	ACB 3P 1250A, 50KA, FT, Ekip Dip LI (*)			ABB, E1.2C12 F/T			3	1Q1-2,2Q1						C	
	11	+ Padlocks For E1.2 (*)			ABB,			3	-							
	12	ACB 3P 1000A, 50KA, FT, Ekip Dip LI (*)			ABB, E1.2C10 F/T			1	2Q2							
	13	+ Padlocks For E1.2 (*)			ABB,			1	-							
	14	MCCB 3P 320-800A, 50KA (*)			ABB, T6S 800 PR221DS-LS/I			1	2Q3							
	15	+ Auxiliary Contact For T5-T6 (*)			ABB, Aux-C 3Q 1SY			1	-							
	16	+ Extended Rotary Handles For T6 (*)			ABB, RHE			1	-							
	17	MCCB 3P 160-400A, 50KA (*)			ABB, T5S PR221DS-LS/I-400			1	1Q3							
	18	+ Auxiliary Contact For T5-T6 (*)			ABB, Aux-C 3Q 1SY			1	-							
D	19	+ Extended Rotary Handles For T5 (*)			ABB, RHE			1	-						D	
	20	Push Button Test Lamp , 1NO, Black			Ega			6	1,2S100-300							
	21	Control Relay 220Vac			Omron, LY-4			6	1,2R1-3							
	22															
	23															
	24															
	25															
	26															
	26															
E																E
F																F
1		2		3		4		5		6		7		8		

				Date Prep.	13.12.2018	PROJECT TITLE		 	COMPONENT LIST		Scale		= H00
				Prepared	S4	LVD for MNA Serang Plant			GA Drawing		Customer	PT MULTIMAS NABATI ASAHAN	+
0	First Issued	13.12.2018		Checked	S1			PT. ABB Sakti Industri	LVD-25A CHILLER SYSTEM FRACTIONATION PLANT		Drawing Nbr.		Sh. B+
Ind.	Revision	Date	Name	Approved	S1	Based on: MNA-BED-RE-DWG-001					Customer Doc. Nbr.		Cont. 21